

Sampat Bandlamudi

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EDUCATION

The University of Texas at Dallas

- B.S. in Data Science | Completed in 3 Years

Richardson, Texas

Aug 2023 - May 2026

Relevant coursework: Machine Learning, Database Systems, Big Data Management, Statistical Learning, Applied Linear Models, Data Structures and Algorithms, Linear Algebra

SKILLS

- **Programming:** Java, Python, SQL, C++, R, JavaScript, PHP, HTML/CSS
- **Frameworks, Cloud & Databases:** React, Node.js, AWS Lambda, API Gateway, DynamoDB, MySQL, Git
- **Software & Data:** REST APIs, CRUD, Data Pipelines, ETL, Data Validation, Database Design, scikit-learn

EXPERIENCE

MITEG – Predictive Oncology Research

Richardson, Texas

- *AI/ML Research Analyst (UT Dallas Industry Capstone with MITEG)*

Aug 2025 – Dec 2025

- Built and evaluated classification models on TCGA-BRCA genomic dataset containing 1,082 patient samples and 13 biomarkers using Python and scikit-learn
- Applied SMOTE, five-fold cross-validation, grid search, and threshold tuning to assess reliability and generalization
- Improved model sensitivity from 61.7% to 78.7% by analyzing classification thresholds and false-negative tradeoffs, with Random Forest achieving the strongest AUROC of 0.682
- Presented methodology, findings, and risk tradeoffs to both the company's CEO and president

RELEVANT PROJECTS

Rainline – AI Irrigation Assistant

- *Project Lead*

Dec 2025 – Mar 2026

- Built a full-stack, serverless irrigation decision platform using React, TypeScript, Node.js, AWS Lambda, DynamoDB, API Gateway, Amazon Bedrock, S3, CloudFront, and AWS CDK
- Developed RESTful API endpoints to process user inputs, retrieve irrigation recommendations, store recommendation history, and capture user feedback
- Created a dual-layer recommendation engine combining rule-based agricultural calculations with AI-generated explanations based on crop type, soil type, growth stage, irrigation history, and weather data
- Recognized as an AWS 10,000 Aldeas National Semi-Finalist

Patient-Link – Clinic Management System

- *Team Lead*

Jun 2025 – Aug 2025

- Led a five-member team developing a PHP and MySQL clinic management system with separate patient, receptionist, and doctor interfaces
- Designed the entity-relationship model, relational schema, functional dependencies, and 3NF-normalized database tables for doctors, patients, appointments, and billing records
- Developed role-based functionality improving workflow efficiency and data organization

Hotel Review Rating Prediction

- *Team Member*

Feb 2025 – May 2025

- Preprocessed data from 3,000+ hotel bookings, performing parsing error correction, missing value imputation, and feature engineering to produce 2,705 analysis-ready observations
- Developed an OLS regression model using AIC-based stepwise feature selection and validated performance with VIF analysis, residual diagnostics, and Cook's distance, explaining 74.6% of the variance in hotel ratings

ADDITIONAL INFORMATION

- **Certifications and Awards:** AWS 10,000 Aldeas National Semi-Finalist, Microsoft Programming Using Python